

Pharmacology of Anti- angina Drugs



Introduction

- *Angina* (sometimes called angina pectoris) is chest pain that is caused by **inadequate coronary blood flow** to the myocardium.
 - When coronary **blood flow cannot deliver** sufficient oxygen to support cardiac oxidative metabolism (reduced oxygen supply/demand ratio), the myocardium becomes **hypoxic**.
 - This triggers **pain receptors** within the heart
 - **classical presentation of chest pain**
 - **sudden, severe pressure**

Introduction

- ❖ Angina pectoris has three overlapping patterns:
 - ✓ effort-induced, stable, classical, or typical angina;
 - ✓ unstable angina; and
 - ✓ Prinzmetal, variant, vasospastic, or rest angina.
- ❖ They are caused by varying combinations of
 - increased myocardial demand and decreased myocardial perfusion

Effort-induced angina, classic or stable angina

- ❖ It is the most common form of angina and,
 - called typical angina pectoris.
- ❖ Characteristics
 - shortlasting burning, heavy, or squeezing feeling in the chest.

Effort-induced angina, classic or stable angina

- **Causes**
 - reduction of **coronary perfusion** due to a **fixed obstruction** of a coronary artery produced by **atherosclerosis**.
 - Heart becomes vulnerable to **ischemia** whenever there is increased demand. ...Demand increases during
 - physical activity,
 - emotional excitement, or
 - any other cause of increased cardiac workload.
- ❖ Typical angina pectoris is promptly **relieved by nitroglycerin**

Unstable angina

- ❖ **In unstable angina**, chest pains occur with increased frequency, duration, and intensity and are precipitated by progressively **less effort**

Unstable angina...

- ❖ suggestive of unstable angina
 - Any episode of rest angina longer than 20 minutes,
 - Any new onset of angina,
 - Any increasing (crescendo) angina, and
 - Sudden development of shortness of breath
- ❖ The symptoms are **not relieved by or nitroglycerin**.
- ❖ **requires hospital admission** and more aggressive therapy to prevent death and progression to MI

Prinzmetal, variant, vasospastic or rest-angina

- ❖ Prinzmetal angina is an **uncommon pattern** of episodic angina
- ❖ occurs **at rest** and is due to **coronary artery spasm**.

Prinzmetal, variant, vasospastic or rest-angina

Causes

- ❖ It is caused by decreased blood flow to the heart muscle from the spasm of the coronary artery
- ❖ Patient may have significant **coronary atherosclerosis**
 - but the angina attacks are unrelated to physical activity, heart rate, or blood pressure
- ❖ responds promptly to coronary vasodilators, such as **nitroglycerin** and **CCB**.

Mixed Forms of Angina

- ❖ Patients with advanced coronary artery disease may present with angina episodes
- ❖ **during effort** as well as **at rest**,
 - suggest the presence of a **fixed obstruction** associated with **endothelial dysfunction** and **vasospastic disease**.

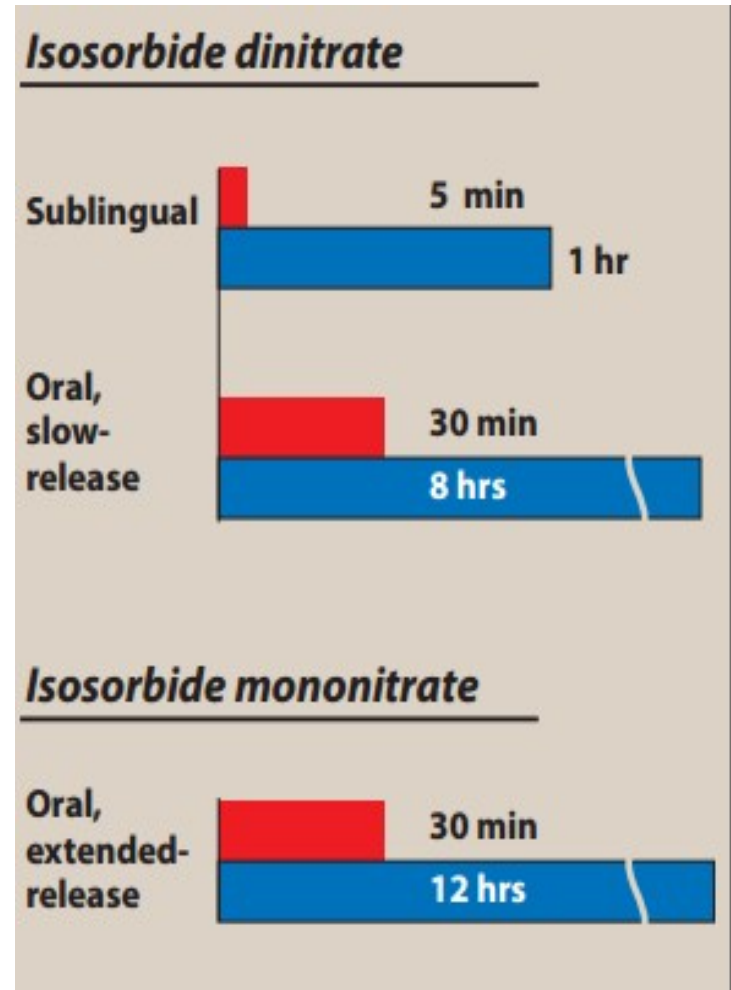
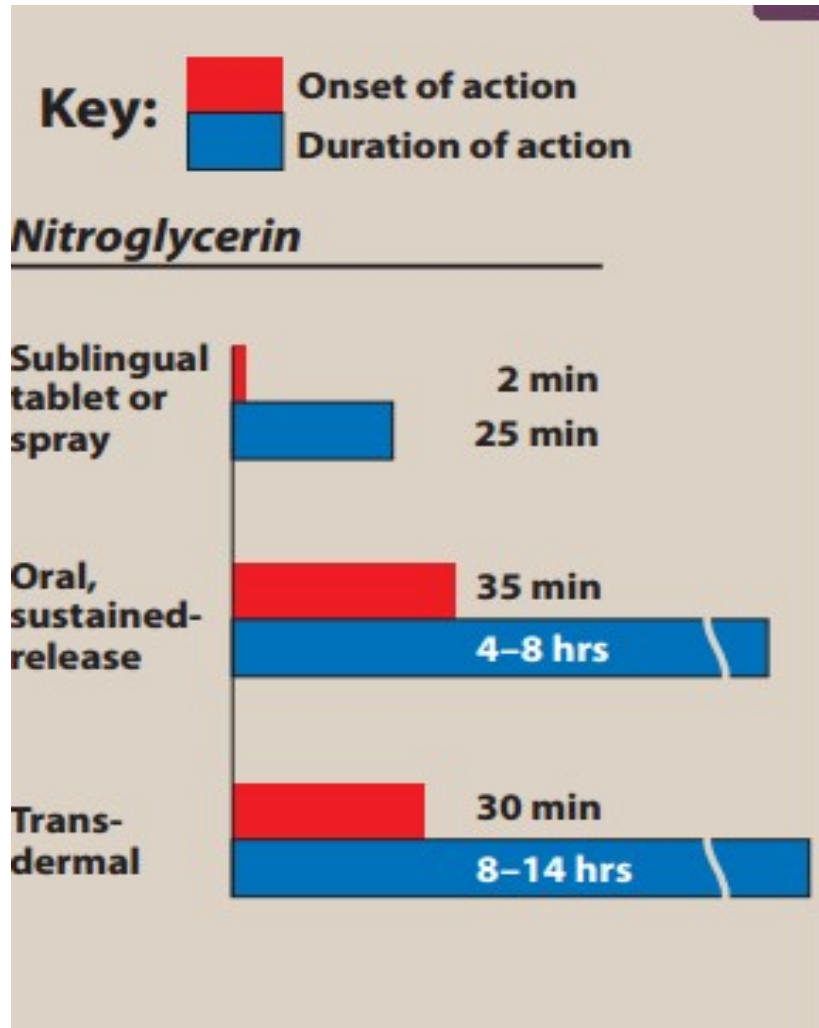
Basic pharmacology of drugs used to treat angina

- ❖ **Vasodilators (dilate arteries and veins)**
 - calcium-channel blockers
 - nitrodilators
- ❖ **Cardioinhibitory drugs (reduce heart rate and contractility)**
 - beta-blockers
 - calcium-channel blockers
- ❖ **Ranolazine (blocker of late sodium currents)**
- ❖ **Anti-thrombotic drugs (prevent thrombus formation)**
 - anticoagulants
 - anti-platelet drugs

Organic nitrates

- ❖ They are simple **nitric** and **nitrous acid esters** of glycerol.
 - They differ in their volatility.
 - isosorbide dinitrate and isosorbide mononitrate..solid
 - Nitroglycerin....moderately volatile, and
 - amyl nitrite....extremely volatile.
- ❖ They cause a **rapid reduction** in myocardial **oxygen demand**, followed by **rapid relief of symptoms**.
- ❖ effective in stable and unstable angina as well as in **variant angina pectoris**

Organic nitrates



Mechanism

- ✓ inhibit coronary spasm..... Increase perfusion
 - relieving vasospastic angina.
- ✓ Produce venodilation, decreasing preload
 - (-)myocardial oxygen consumption
 - effective in treating effort-induced angina (classic angina)

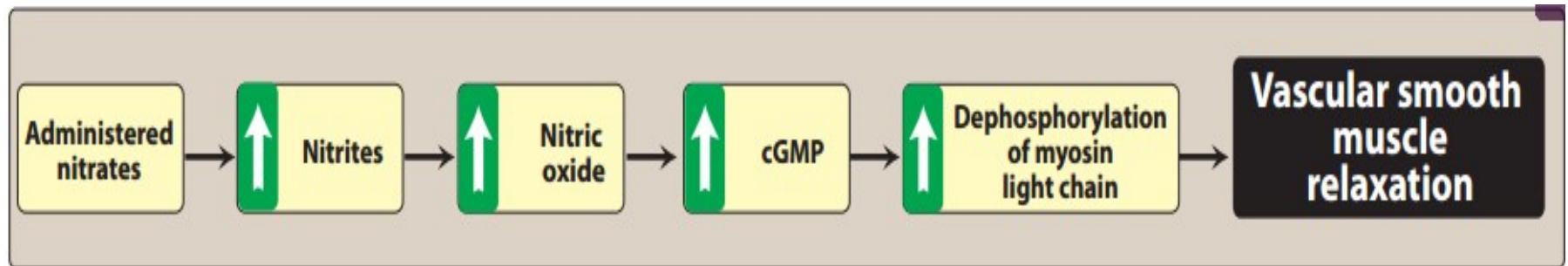


Figure: Effects of nitrates and nitrites on smooth muscle

Effects on the cardiovascular system

- ❖ they differ in their onset of action and rate of elimination
 - an ongoing attack of angina precipitated by exercise or emotional stress,
 - sublingual (or spray form) nitroglycerin is the drug of choice
- ❖ Nitroglycerin has two major effects.
 - First, it causes dilation of the large veins...diminishes preload (venous return to the heart)
 - reduces the work of the heart (decrease consumption)
 - dilates the coronary vasculature, providing an increased blood supply to the heart muscle..

Pharmacokinetics

- ❖ The time to onset of action varies (1 minute - ≥ 1 hr)
- ❖ Significant first-pass metabolism of **nitroglycerin** occurs in the liver.
 - common to take the drug either sublingually or via a transdermal patch,
- ❖ Isosorbide mononitrate owes its improved bioavailability and long duration of action
- ❖ Oral isosorbide dinitratedenitration to two mononitrates,
 - both of which possess antianginal activity

Adverse effects

- ❖ The most common adverse effect, is headache (30-60 %)
- ❖ High doses of organic nitrates can also cause postural **hypotension**,
 - C/I.....with Phosphodiesterase V inhibitors
- ❖ Facial flushing

❖ Tolerance to the actions of nitrates develops rapidly.

–The blood vessels become desensitized to vasodilation.

Solution

–overcome by providing a daily “nitrate-free interval” to restore sensitivity to the drug.

- This interval is typically 10 to 12 hours, usually at night, because demand **(-) during that time .**

–Nitroglycerin patches are worn for 12 hours, then removed for 12 hours.

–variant angina worsens early in the morning, perhaps due to circadian catecholamine surges.

–**nitrate-free interval** in these patients should occur in the late afternoon

–Patients who continue to have angina despite nitrate....Add other class of drug

- ❖ Exposure to high levels of nitrates In manufacturing work place with volatile organic nitrate results in headache and transient dizziness
- ❖ This condition is known to be “Monday disease “
why?

B-adrenergic Blockers

- ❖ These drugs decrease the oxygen demands by lowering
 - the **rate** and the **force of contraction**
 - cardiac output, and
 - blood pressure
- ❖ reduce demand both **during exertion** and **at rest**
 - the drugs of choice to treat **effort-induced angina**
- ❖ They reduce the **frequency** and **severity** of angina attacks
- ❖ ineffective **against** and **should not** be used in **vasospastic angina**.

Calcium-channel blockers

- ❖ Calcium is essential for muscular contraction.
- ❖ Calcium influx is increased in ischemia
 - because of the membrane depolarization that hypoxia produces.
 - ✓ this promotes the activity of several ATP – consuming enzymes
 - ✓ depleting energy stores and worsening the ischemia
- ❖ **CCBs** protect the tissue by inhibiting the entrance of calcium into cardiac and smooth muscle cells of the coronary and systemic arterial beds (**produce arterial dilation**).....Cause a reduction in vascular resistance

Calcium-channel blockers...

- ❖ At clinical doses, CCBs
- (-) resistance of peripheral and coronary arteriolar smooth muscle.

Role of CCBs in Angina

- ❖ **effort-induced angina**
 - relies on the reduction in **myocardial oxygen consumption** resulting from **decreased afterload**. Their efficacy
- ❖ **vasospastic angina**
 - due to relaxation of the coronary arteries.
- ❖ Verapamil mainly **affects** the myocardium,
- ❖ Nifedipine exerts a greater effect on smooth muscle in the peripheral vasculature.
- ❖ Diltiazem is intermediate in its actions

Calcium-channel blockers...

❖ **Variant angina** caused by spontaneous **coronary spasm**, either at work or at rather than by **increased myocardial oxygen requirement**,

—is controlled by **organic nitrates** or **calcium-channel blockers**.

• β -Blockers are contraindicated

Nifedipine

❖ has minimal effect on **cardiac conduction** or **heart rate**.

❖ Given orally, usually as **extended-release tablet** (**Short acting should be avoided**)

—— useful in the treatment of variant angina caused by spontaneous coronary spasm

S/Esflushing, headache, hypotension, and peripheral edema as side effects of its vasodilation activity, Constipation & Reflex Tachycardia

Verapamil

- ❖ It is a diphenylalkylamine derivative

Effect on the Heart and Blood Vessels

- ❖ slows cardiac atrioventricular (AV) Conduction
- ❖ decreases heart rate, contractility, blood pressure, and oxygen demand.
- ✓ It causes greater **negative inotropic effects** than nifedipine, but it is a **weaker vasodilator**.
- ✓ extensively metabolized by the liver,
 - dose should be adjusted in patients with liver dysfunction.

C/I

- in patients with preexisting **depressed cardiac function** or **AV conduction abnormalities**

Verapamil

- ❖ causes constipation.
- ❖ Verapamil + Digoxin
 - verapamil increases digoxin levels

Diltiazem

- ❖ Can relieve coronary artery spasm...
 - particularly useful in patients with **variant angina**.
- ❖ has cardiovascular effects that are similar to those of verapamil.
- ❖ Reduces heart rate (to lower extent than verapamil)
- ❖ Drug interaction....The same to verapamil

Sodium-channel blocker

- ❖ Ranolazine inhibits the late phase of the sodium current (late I_{Na})
 - Improve diastolic function.

Role in angina

- for the treatment of chronic angina and may be used alone or in combination
 - often used as an option in angina patients who have failed all other antianginal therapies.
- ❖ It is **not to be used to treat an acute attack of angina.**

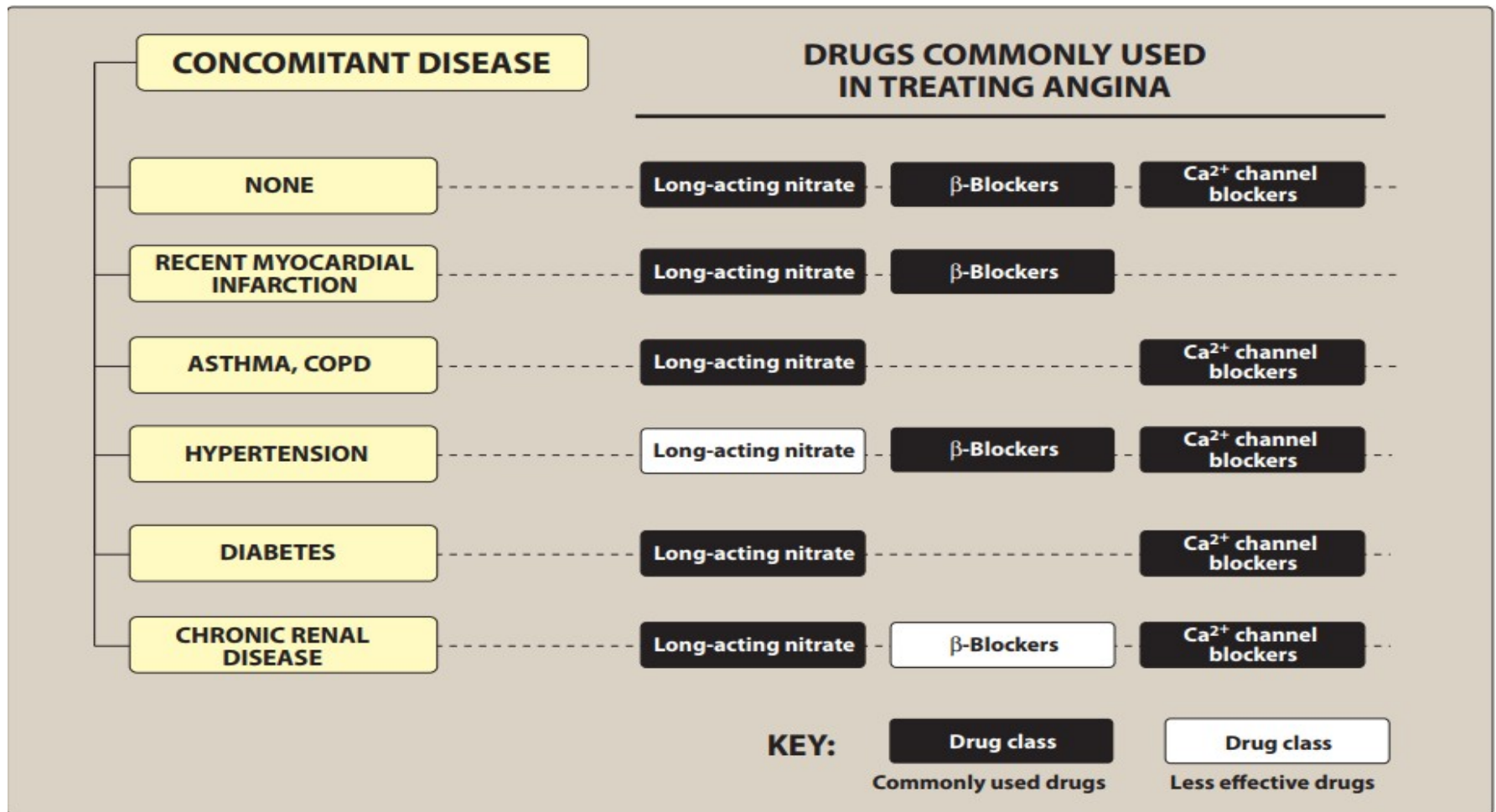


Figure: Treatment of angina in patients with concomitant diseases